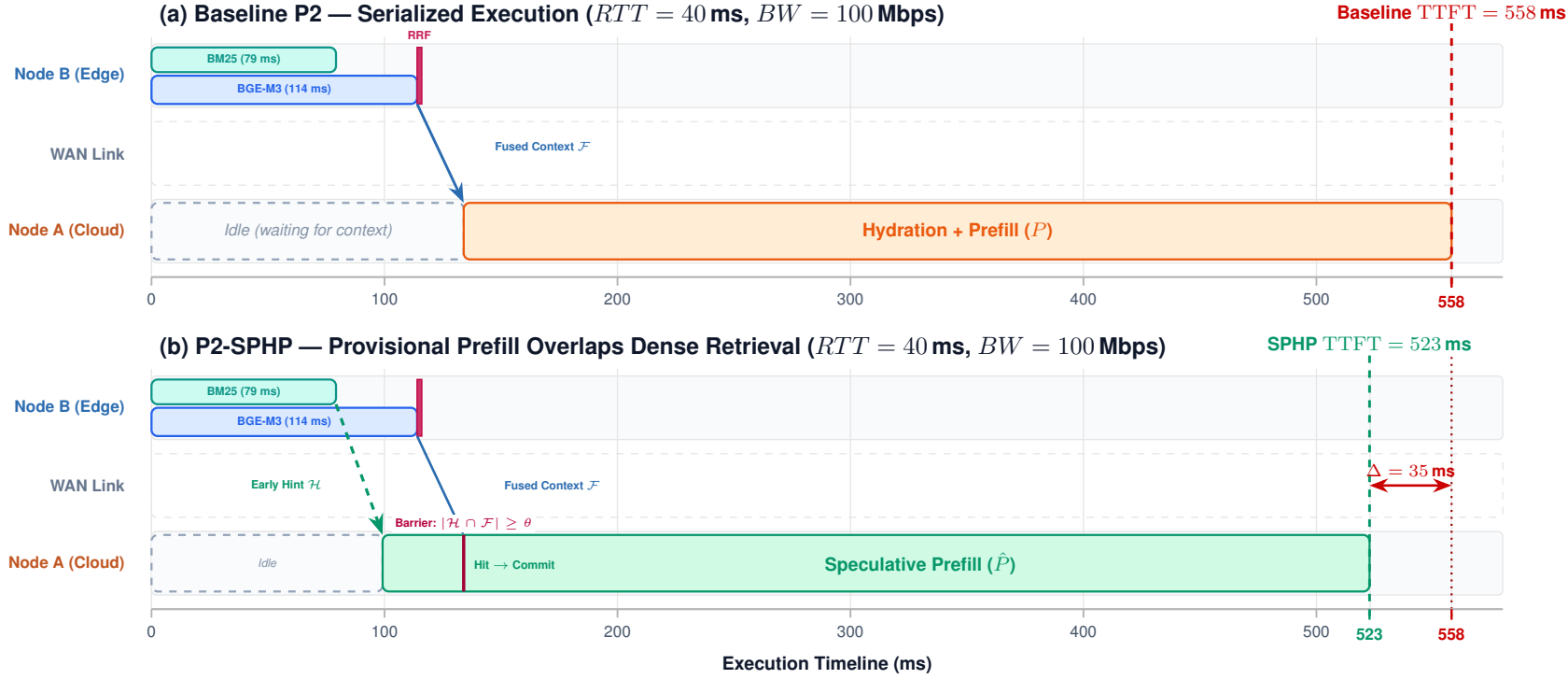

(a) Baseline P2 — Serialized Execution ($RTT = 40$ ms, $BW = 100$ Mbps)
Baseline TTFT $= 558$ ms
RRF
Node B (Edge)
BM25 (79 ms)
BGE-M3 (114 ms)
WAN Link
Fused Context $\mathcal{F}$
Node A (Cloud)
Idle (waiting for context)
Hydration + Prefill (P)
0
100
200
300
400
500
558
(b) P2-SPHP — Provisional Prefill Overlaps Dense Retrieval ($RTT = 40$ ms, $BW = 100$ Mbps)
SPHP TTFT $= 523$ ms
Node B (Edge)
BM25 (79 ms)
BGE-M3 (114 ms)
WAN Link
Early Hint $\mathcal{H}$
Fused Context $\mathcal{F}$
$\Delta = 35$ ms
Barrier: $|\mathcal{H} \cap \mathcal{F}| \geq \theta$
Node A (Cloud)
Idle
Hit → Commit
Speculative Prefill ($\hat{P}$)
0
100
200
300
400
500
523
558
Execution Timeline (ms)